

Lecture 25: Surface Integrals

Background & Motivation

We've integrated functions along curves (line integrals) and over flat regions (double integrals). Today we integrate over curved surfaces in 3D. Surface integrals compute mass of a shell, and flux integrals measure how much a vector field flows through a surface. These are the ingredients we need for Stokes' and the Divergence Theorem.

1. Parametric Surfaces

Definition 1: Parametric Surface

A **parametric surface** S is the image of a continuous map $\mathbf{r} : D \rightarrow \mathbb{R}^3$, where $D \subseteq \mathbb{R}^2$. Each pair $(u, v) \in D$ is sent to a point on S — think of latitude and longitude on a globe.

$$\mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle, \quad (u, v) \in D.$$

- **Tangent vectors:** $\mathbf{r}_u = \frac{\partial \mathbf{r}}{\partial u}$, $\mathbf{r}_v = \frac{\partial \mathbf{r}}{\partial v}$
- **Normal vector:** $\mathbf{n} = \mathbf{r}_u \times \mathbf{r}_v$ (perpendicular to S ; nonzero where S is smooth)
- **Surface area element:** $dS = \|\mathbf{r}_u \times \mathbf{r}_v\| du dv$

Common Parameterizations

Surface	Parameterization $\mathbf{r}(u, v)$	Domain	$dS = \ \mathbf{r}_u \times \mathbf{r}_v\ du dv$
Graph $z = g(x, y)$	$\langle x, y, g(x, y) \rangle$	$(x, y) \in D$	$\sqrt{1 + g_x^2 + g_y^2} dA$
Sphere $x^2 + y^2 + z^2 = a^2$	$\langle a \sin \phi \cos \theta, a \sin \phi \sin \theta, a \cos \phi \rangle$	$\theta \in [0, 2\pi], \phi \in [0, \pi]$	$a^2 \sin \phi d\phi d\theta$
Cylinder $x^2 + y^2 = R^2$	$\langle R \cos \theta, R \sin \theta, z \rangle$	$\theta \in [0, 2\pi], z \in [c, d]$	$R d\theta dz$
Cone $z = \sqrt{x^2 + y^2}$	$\langle r \cos \theta, r \sin \theta, r \rangle$	$\theta \in [0, 2\pi], r \in [0, h]$	$r\sqrt{2} dr d\theta$
Plane thru P_0 , dirs $\mathbf{a}, \mathbf{b}$	$P_0 + s\mathbf{a} + t\mathbf{b}$	$(s, t) \in D$	$\ \mathbf{a} \times \mathbf{b}\ ds dt$
Rev. of $y = f(x)$ about x -axis	$\langle x, f(x) \cos \theta, f(x) \sin \theta \rangle$	$x \in [a, b], \theta \in [0, 2\pi]$	$ f(x) \sqrt{1 + f'(x)^2} dx d\theta$

Special Case: $z = g(x, y)$

$$dS = \underline{\hspace{2cm}}$$

$$\mathbf{n} dS = \langle -g_x, -g_y, 1 \rangle dA \quad (\text{upward-pointing normal})$$

For downward normal, negate: $\langle g_x, g_y, -1 \rangle dA$.

2. Scalar Surface Integrals

Definition 2: Surface Integral of a Scalar Function

The **surface integral** $\iint_S f \, dS$ generalizes the double integral from a flat 2D region to a curved surface in $\mathbb{R}^3$. It sums values of f over S weighted by the true area element dS .

$$\iint_S f \, dS = \underline{\hspace{2cm}}$$

For $z = g(x, y)$:

$$\iint_S f \, dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2} \, dA$$

– **Applications:** mass ($f = \rho$), average value on a surface

Special Case: $f \equiv 1$ gives Surface Area

Setting $f = 1$ in the surface integral turns it into the **surface area** of S :

$$\text{Area}(S) = \iint_S 1 \, dS = \underline{\hspace{2cm}}$$

For a graph $z = g(x, y)$ over a region D in the xy -plane:

$$\text{Area}(S) = \iint_D \sqrt{1 + g_x^2 + g_y^2} \, dA$$

Example 1: Mass of a Hemispherical Shell

Find the mass of the upper hemisphere $x^2 + y^2 + z^2 = 9$, $z \geq 0$, with density $\rho = z$.

3. Flux Integrals

The **flux** of a vector field $\mathbf{F}$ across an oriented surface S measures the net rate at which $\mathbf{F}$ crosses S in the direction of the chosen unit normal $\hat{\mathbf{n}}$. If $\mathbf{F}$ is a fluid velocity field, the flux is the volume of fluid passing through S per unit time.

For $z = g(x, y)$ with **upward** normal:

– Interpretation: _____

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- A diagram illustrating a curved surface S with a grid of lines. A vector field F is shown as red arrows originating from various points on the surface. A specific point on the surface is labeled p , and the vector at this point is labeled dF , shown as a blue arrow. The surface is shaded gray.

Example 2: Flux Through a Paraboloid

Find the flux of $\mathbf{F} = \langle y, x, z \rangle$ through the paraboloid $z = 1 - x^2 - y^2$, $z \geq 0$, with upward normal.

Example 3: Flux Through a Sphere

Find the outward flux of $\mathbf{F} = \langle x, y, z \rangle$ through the sphere $x^2 + y^2 + z^2 = a^2$.

Practice Problems

Problem 1. Find the surface area of the plane $z = 3 - x - y$ over the triangle $0 \leq x \leq 1$, $0 \leq y \leq 1 - x$.

Problem 2. Find the flux of $\mathbf{F} = \langle 0, 0, 1 \rangle$ upward through $z = 1 - x^2 - y^2$ above $z = 0$.

Problem 3. Compute $\iint_S z \, dS$ where S is the cone $z = \sqrt{x^2 + y^2}$, $0 \leq z \leq 2$.

(Hint: $g_x = x/\sqrt{x^2 + y^2}$, $g_y = y/\sqrt{x^2 + y^2}$, so $\sqrt{1 + g_x^2 + g_y^2} = \sqrt{2}$.)

Problem 4. Find the flux of $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$ outward through the unit sphere $x^2 + y^2 + z^2 = 1$.
(Hint: on the unit sphere the outward unit normal is $\hat{\mathbf{n}} = \langle x, y, z \rangle$, so $\mathbf{F} \cdot \hat{\mathbf{n}} = x^3 + y^3 + z^3$. Use symmetry.)

Problem 5. Find the flux of $\mathbf{F} = \langle z, x, y \rangle$ upward through the triangle cut from the plane $x + y + z = 1$ in the first octant.